CHMENG 188/288 Electrochemical Projects Laboratory
Summer Session D 2025 (6 weeks)
Transcript Title: Electrochem Proj

I. Course Identity, Teaching Staff, and Logistics

Instructor: Prof. Shannon Boettcher, email: boettcher@berkeley.edu, Office: 212 Gilman

This is a new team-project based laboratory course that will be taught/co-taught by Prof. Boettcher, Prof. Bryan McCloskey, Prof. Niel Razdan, Prof. Nitash Balsara, PhD level Lawrence Berkeley National Lab Scientists, and/or a new electrochemistry instructor / program director (PhD level, TBD).

Graduate Student Instructor (GSI): Student Name will serve as a GSI for this course. They are available at GSIemail@berkeley.edu and will work with student teams to guide the team-based research projects.

Office hour: There will be two office hours per week. The times will be determined on the first day of lecture based on student availability.

Format: Laboratory and team-based learning.

Credits: 3

Location: The team research project will make use of both teaching laboratory and specialized research laboratory space and equipment. Required laboratory hours are 1-5 pm four days per week (M, Tu, W, Th).

Required course materials:

Electrochemistry Projects Laboratory Manual: Each student will be provided a laboratory manual that includes key articles from the scientific literature related to the team's research project. Typically these will include both review/tutorial articles (20-50 journal pages in length) and 5-10 key original research articles. Relevant book chapters will also be assigned for reading.

Additional course materials:

The following texts are used in the lecture components of the Berkeley summer electrochemistry academy. They are not required for this course, but students will find the content useful in electrochemistry research projects.

Fuller and Harb*. Electrochemical Engineering,* Wiley 2018. ISBN-13: 978-1119004257

Oldham, Myland, and Bond. *Electrochemical Science and Technology.* Wiley 2012.
(available as free PDF download for UCB):
https://onlinelibrary.wiley.com/doi/pdf/10.1002/9781119965992)

Course website: Laboratory Project Assignments, lecture notes, videos, and grades will be posted on the course site.

Prerequisites: CHEMENG 186/286 Electrochemistry Fundamentals

II. Course Description

Students work in teams to solve open-ended research and development projects in electrochemical science, engineering, and technology. The projects for the course come from industry partners, national laboratory partners, and academic research laboratories. This allows the students to develop skills solving unstructured problems representative of what they will face in their career. Example projects span electrolysis and fuel cells, interfacial electrochemistry, batteries, and electrosynthesis.

Additional Description

The output will be a formal report or academic peer-reviewed publication where the students are co-authors. The class will meet four afternoons a week for mandatory laboratory/group-discussion sessions led by the instructor and/or teaching assistant. During this time the teams will present their weekly progress update, be able to ask questions of the instructor(s), and receive training for any equipment as needed. Open laboratory times will also be established for students to work at their own pace and on their own schedule on the assigned projects, after appropriate safety training and with oversight as needed. Feedback from industry partners will drive laboratory project content evolution. Experienced PhD student graduate student instructors (GSIs) with expertise in the project content will mentor and work with the teams.

III. Expected Learning Outcomes

- Develop project planning and management skills needed to complete longer scale applied research and development projects
- Develop presentation and speaking skills necessary to give routine technical project updates
- Learn to apply foundational concepts in chemical thermodynamics, kinetics, and mass transport to optimize the operation and performance of practical electrochemical devices
- Develop deep understanding of specific electrochemical device being researched

- Learn to find and assess the primary literature to drive decision making

IV. Estimated Student Workload

Course participants will attend laboratory sessions, perform assigned reading, attend open laboratory times and collect data, analyze data, present their data, and write a final professional report or manuscript for submission to a peer reviewed journal. The table below shows the estimated workload.

Activity	Estimated hours per term	Comments
Preparation for weekly project updates	10	5 project updates, 2 h per update
Project Literature Readings	18	Estimated at 3 h / week
Laboratory Sessions	96	16 h / week
Final report/paper	10	High quality report or paper suitable for submission to peer-reviewed journal. Each team (~3 students) will work in collaboration to draft paper.
Total hours:	134	Consistent with 3 credit workload

V. How Grades Will Be Determined

The grades will be determined based on the following percentage breakdown of the final total score:

<u>Graduate CHMENG 288 Students</u>

Professionalism and safety in the laboratory setting - 10%
Quality of laboratory notebook data collection and analysis – 25%
Quality of weekly research updates – 25%
Final project report/paper – 25%
Independent lecture on chemistry, physics, and engineering concepts underlying research project – 15%

<u>Undergraduate CHMENG 188 Students</u>

Professionalism and safety in the laboratory setting - 10%
Quality of laboratory notebook data collection and analysis – 30%

Quality of weekly research updates – 30%
Final project report/paper – 30%

Professionalism and safety in the laboratory setting will be assessed by the graduate student instructor and instructor of record throughout the course. Students engaging in unsafe laboratory practices will lose credit and be given a warning. A second infraction will result in the student being removed from the course. Students are also expected to interact with each other in a professional manner with effort focused and meeting the project deliverables. Unprofessional conduct will result in loss of credit.

The quality of each student's laboratory notebook (digital and physical) will be assessed through periodic unannounced notebook checks. During a check each student must be able to find and explain key data collected, including providing details of how the data was collected and the sample/device history. The goal of the notebook checks is to teach appropriate attention to detail when recording experimental data and observations.

Each student will also give a weekly update on research progress (10 min with questions). Each presentation should be concise and clear. Each presenter will answer questions pertaining to the presentation. Feedback will be given after each presentation so the presenter can systematically improve their presentation skills during the course of the 10-week class. Grades will be based on the clarity of the presented data and quality of the information delivery.

The final project/paper will be graded based on the quality of the document. Reports/papers are expected to be concise formal documents (if a manuscript it must be formatted to meet ACS or other publication standard) with professional quality graphs (including appropriate use of units and error estimates) and data analysis, formatted references, and insightful discussion of the data. Students will receive a separate document detailing expectations.

Graduate students enrolled in the CHMENG 288 course will be separately required to make a pedagogical technical presentation on the underlying chemistry, physics, and engineering concepts governing their research project.

A = Clearly written, concise document, with high-quality figures. All required pieces of data are shown and discussed without major errors in the interpretation. Appropriate use of references.

B = One or more issues with the items above.

C = Multiple issues with quality of the written document and interpretation/discussion.

D = Multiple issues and incomplete.

+/- grades will be applied consistent with the above criteria. A+ will be given for near-perfect reports with high quality data.

Students earning >90% in the course will earn at least and A, > 80% at least a B, >70% at least a C, and >60% at least a D. The instructor may curve the course to increase the letter grades for a given percentage score to account for variations in difficulty of the exam questions from year-to-year. Students with a given percentage score will not, however, earn a grade lower than that indicated above.

VI. Course Schedule and Assignments

Below is an example of curriculum for one of the team-projects based on advanced analysis and design of water electrolyzers to support industry decarbonization goals. The content below is specific to this example team.

Week 1

Laboratory Meeting 1: Laboratory Safety Review. Training on key electrochemical instrumentation, power supplies, potentiostats, and the control and measurement of current and voltage. Project descriptions and assignments.

Laboratory Meeting 2: Project selection and individual team project discussions and planning. Example breadth of team mentor/guided research projects include: 1) durability and catalytic mechanisms of electrodes in advanced alkaline water electrolysis for hydrogen production; 2) reversible Zn electrodeposition and connection between electrode structure and faradaic efficiencies, finite element modeling of membrane electrode assemblies, 3) high current advanced electrodialysis for carbon capture, 4) CO_2 electrolyzer cell design and testing, conversion electrodes for next-generation energy high energy density energy storage.

Reading: Primary literature associated with team's specific research project. Example:

Varcoe, J. R., et al. (2014). "Anion-exchange membranes in electrochemical energy systems." *Energy & Environmental Science* 7, 3135-3191.

Pivovar, B. S., et al. (2019). "Hydroxide exchange membrane fuel cells and electrolyzers: Status and challenges." *Journal of The Electrochemical Society* 166, F1024-F1031.

Laboratory Meeting 3: Weekly group project updates from each team (<10 min per team). Team learns how to make measurements of electrocatalytic performance on thin film electrodes and measures baseline catalysts. Techniques from: Burke, M. S., et al. (2015). "Oxygen evolution reaction electrocatalysis on transition metal oxides and (oxy)hydroxides: Activity trends and design principles." *Chemistry of Materials* 27(22), 7549-7558.

Laboratory Meeting 4: Quantifying and understanding catalytic activity; Tafel slope, electrochemical surface area, exchange current, and turn-over frequency.

Reading: Primary literature associated with team's specific research project.

Coutard, N., et al. (2017). "Development of stable catalysts for oxygen evolution in alkaline media." *Nature Communications* 8, 15438

Week 2

Laboratory Meeting 1: Weekly project planning report from each team (5-10 min per team). Continue catalyst measurement and benchmarking of new catalyst compositions/architectures versus baseline.

Laboratory Meeting 2: Instrumentation and methods for quantifying durability and corrosion processes in electrochemical cells. Learn about and practice electrochemical quartz crystal microbalance measurements on active electrocatalysts.

Reading: Primary literature associated with team's specific research project.

Deakin, M. R. & Buttry, D. A. Electrochemical applications of the quartz crystal microbalance. *Anal. Chem.* **61**, 1147A-1154A (1989).

Laboratory Meeting 3: Catalyst ink preparation. Learn about the role of ionomer-catalysts interactions in stabilizing ink dispersion.

Reading: Berlinger, S. A., McCloskey, B. D. & Weber, A. Z. Probing Ionomer Interactions with Electrocatalyst Particles in Solution. *ACS Energy Lett.* **6**, 2275-2282 (2021).

Laboratory Meeting 4: Weekly group project updates from each team (<10 min per team). Laboratory research with team on catalyst ink, preparation, testing, and data analysis in Python.

Week 3

Laboratory Meeting 1: Weekly project planning report from each team (5-10 min per team). Training on preparing membrane electrode assemblies (MEAs) including catalyst layers, ionomers, and membranes.

Reading: Primary literature associated with team's specific research project.
Bhosale, A. C., Ghosh, P. C. & Assaud, L. Preparation methods of membrane electrode assemblies for proton exchange membrane fuel cells and unitized regenerative fuel cells: A review. *Renewable and Sustainable Energy Reviews* **133**, 110286 (2020).

Laboratory Meeting 2: Weekly project updates from each team. Team prepares baseline and test membrane electrolyzer cells using standard baseline and new catalyst powders.

Reading: Primary literature associated with team's specific research project.
Lindquist, G. A., Xu, Q., Oener, S. Z., & Boettcher, S. W. (2020). "Membrane electrolyzers for impure-water splitting." *Joule*, 4, 2549-2561.

Laboratory Meeting 3: Weekly project planning report from each team (5-10 min per team). Team electrolyzer testing and data analysis. Measuring and quantifying electrolyzer

performance metrics.

Laboratory Meeting 4: Team electrolyzer testing and data analysis. Measuring and quantifying electrolyzer durability metrics.

Reading: Lindquist, G. A., Oener, S. Z., Krivina, R., Motz, A. R., Keane, A., Capuano, C., Ayers, K. E. & Boettcher, S. W. Performance and Durability of Pure-Water-Fed Anion Exchange Membrane Electrolyzers Using Baseline Materials and Operation. *ACS Appl. Mater. Interfaces* **13**, 51917-51924 (2021).

Week 4

Laboratory Meeting 1: Weekly project planning report from each team (5-10 min per team). Understanding modes of activation and degradation in dynamic electrolyzer electrodes.

Reading: Lindquist, G. A., Gaitor, J. C., Thompson, W. L., Brogden, V., Noonan, K. J. T. & Boettcher, S. W. Oxidative instability of ionomers in hydroxide-exchange-membrane water electrolyzers. *Energy Environ. Sci.* **16**, 4373-4387 (2023).

Laboratory Meeting 2: Continued electrolyzer device fabrication and device testing

Laboratory Meeting 3: Weekly project planning report from each team (5-10 min per team). Continued electrolyzer device fabrication and device testing. Voltage breakdown analysis of polarization curves.

Reading: Gerhardt, M. R., Pant, L. M., Bui, J. C., Crothers, A. R., Ehlinger, V. M., Fornaciari, J. C., Liu, J. & Weber, A. Z. Method—Practices and Pitfalls in Voltage Breakdown Analysis of Electrochemical Energy-Conversion Systems. *J. Electrochem. Soc.* **168**, 074503 (2021).

Laboratory Meeting 4: Continue device fabrication with different materials sets (ionomer polymers, catalysts, layer thicknesses etc) along with quantitative voltage breakdown analysis of polarization curves.

Week 5

Laboratory Meeting 1: Weekly project planning report from each team (5-10 min per team). Device fabrication and down selection. Impedance analysis of membrane electrolyzers.

Reading: Watzele, S. A., Kluge, R. M., Maljusch, A., Borowski, P. & Bandarenka, A. S. Impedance Response Analysis of Anion Exchange Membrane Electrolyzers for Determination of the Electrochemically Active Catalyst Surface Area. *Chemistry–Methods* **4**, e202300035 (2024).

Laboratory Meeting 2: Post mortem electrolyzer tear down and analysis. Plan final experiments for paper and presentation.

Reading: Glenn, J. R., Lindquist, G. A., Roberts, G. M., Boettcher, S. W. & Ayers, K. E. Standard

operating procedure for post-operation component disassembly and observation of benchtop water electrolyzer testing. *Front. Energy Res.* **10**, 10.3389/fenrg.2022.908672 (2022).

Laboratory Meeting 3: Weekly project planning report from each team (5-10 min per team). Continue electrolyzer tear-down and analysis.

Laboratory Meeting 4: Final experimental designs due, team down selects key questions for final set of data to evaluate.

Week 6

Laboratory Meeting 1: Teams fabricate final devices, collect catalyst layer, membrane, ionomer characterization data including polarization, impedance, tear-down analysis to test performance durability and operating modes of selected new materials / catalyst / engineering design approach.

Laboratory Meeting 2: Teams continue testing and prepare data analysis for final project presentation and report.

Laboratory Meeting 3: Final project team presentations.
Laboratory Meeting 4: Final project team presentations.
Final project/paper due.

VII. Course Policies

- Late or missed work will not generally be accepted without prior approval.
- Project reports are collaborative efforts. But all students will be responsible for understanding the research data collected and subsequent analysis and contributing to the final project paper and presentation. If it appears some student are not meeting these requirements they may be asked to participate in a private oral examination on the topic of their research project to assess their understanding and contributions.

Academic Misconduct: The University Student Conduct Code (available at https://conduct.berkeley.edu/code-of-conduct/) defines academic misconduct. Students are prohibited from committing or attempting to commit any act that constitutes academic misconduct. By way of example, students should not give or receive (or attempt to give or receive) unauthorized help on assignments or examinations without express permission from the instructor. Students should properly acknowledge and document all sources of information (e.g. quotations, paraphrases, ideas) and use only the sources and resources authorized by the instructor. If there is any question about whether an act constitutes academic misconduct, it is the students' obligation to clarify the question with the instructor before committing or attempting to commit the act.

Use of Artificial Intelligence, ChatGPT, and other related technologies:

AI is a powerful tool to help you learn and improve your communication. You are welcome to make use of AI tools in this class. However, directly copying from AI sources is considered plagiarism. AI written text is also poor and use of AI without applying your own critical thinking and editing skills will likely result in a poor grade. AI may also be helpful in performing calculations. However, each student is responsible for understanding the calculations and analysis as well as the language used in their writing. Any student may be asked to explain concepts and repeat calculations in person if there is suspected plagiarism from AI work or from other humans.

Statement on Classroom Climate:
We are all responsible for creating a learning environment that is welcoming, inclusive, equitable, and respectful. The expectation in this class is that we all live up to this responsibility, even during vigorous debate or disagreement, and that we will intervene if exclusionary or harassing behavior occurs. If you feel that these expectations are not being met, you can consult your instructors or seek assistance from campus resources.

Academic Accommodations and Accessibility:
The purpose of academic accommodations is to ensure that all students have a fair chance at academic success. If you have Letters of Accommodations from the Disabled Students' Program or another authorized office, please share them with me as soon as possible, and we will work out the necessary arrangements. While individual circumstances can vary, requests for accommodations often fall into the categories listed on the Academic Calendar and Accommodations website. The campus has well-developed processes in place for students to request accommodations, and you are encouraged to contact the relevant campus offices listed on the Academic Accommodations Hub. These offices, some of which are confidential, can offer support, answer questions about your eligibility and rights, and request accommodations on your behalf, while maintaining your privacy.

Electrochemistry Science, Engineering, and Technology Course Sequence Project and Laboratory Report Grading Rubric

	4 - Good	3 – Barely/nearly acceptable	2 - Needs improvement	1 - Not acceptable	0 – Bad	SCORE
Introduction	A cohesive, but concise, well-written summary (including all relevant chemistry, physics, and engineering concepts) of the background material pertinent to the experiment with appropriate references. Places the purpose of the experiment in context.	Is nearly complete but does not provide context for minor points. Contains relevant information but fails to provide background for one aspect of the project or certain information is not cohesive. Certain major introductory points are missing (ex: background, theory, context, etc.) or explanations are unclear and confusing. References are used properly.	Certain major introductory points are missing (ex: background, theory, chemistry, context, etc.) or explanations are unclear and confusing. References are used properly but are not complete.	Very little background information is provided and/or information is incorrect. Poor use of references.	None or unrelated	_____
Experimental	Contains details on how the experiment/simulation was performed and the procedures followed. The experimental setup must be described, detailing the instrumentation/software used and what purpose each piece of equipment serves in the overall experiment. This information should be presented in a brief and concise manner. Written grammatically correctly and omits information that can be assumed by peers (electrochemists)	Narrative includes most important experimental detail but is missing one or more relevant pieces of information.	Missing several experimental details or some incorrect statements.	Several important experimental details are missing. Narrative is incorrect, illogical, or copied directly from other sources. Written in the incorrect tense.	None or unrelated	_____
Results Including experimental data, provided project data, or data from numerical simulation	All figures, graphs, and tables are numbered with appropriate captions and of professional quality. All units are provided for data. All tables, figures, etc. are explicitly mentioned in the text. Relevant experimental data are presented which are used in the discussion.	All figures, graphs, and tables are correctly drawn, but some have minor problems that could be still be improved. Units are used consistently. Figures quality is good, but could be improved. All data and associated figures, etc. are mentioned in the text. Most relevant data present.	Most figures, graphs, and tables are included, but some important or required features are missing. Certain data reported are not mentioned in the text or are missing. Captions are not descriptive or incomplete.	Figures, graphs, and tables are poorly constructed; have missing titles, captions or numbers. Certain data reported are not mentioned in the text. Important data missing.	None or unrelated	_____ × **2**

Discussion/ Conclusions	Demonstrates a logical, coherent working knowledge and understanding of important concepts, forms appropriate conclusions based on interpretations of results, includes references, data and analysis, refers to the literature when appropriate, and demonstrates accountability by providing justification for any errors. Key equations are justified and their application appropriately discussed. All equations are appropriately typeset. Address all specific points or questions posed in the project/laboratory guidelines/manual.	Demonstrates an understanding of the majority of important concepts, forms conclusions based on results and/or analysis but either lacks complete interpretation, refers to the literature insufficiently, or lacks complete justification of error and or use of models and theoretical equations. Address most of the specific points or questions posed in the project/laboratory guidelines/manual.	While some of the results have been correctly interpreted and discussed, partial but incomplete understanding of results is evident. Student fails to make one or two connections to underlying theory. Address only some of the specific points or questions posed in the project/laboratory guidelines/manual.	Does not demonstrate an understanding of the important experimental concepts, forms inaccurate conclusions, suggests refers to the literature insufficiently, and lacks overall justification of error and/or use of models and theoretical equations. Address none of the specific points or questions posed in the project/laboratory guidelines/manual.	None or unrelated	____ × **2**
References	All sources (information and graphics) are accurately documented in ACS or AIP format.	All sources are accurately documented, but formatting is not consistent or always correct.	Some sources are not accurately documented. Formatting is not consistent or correct.	Sources are not accurately documented. Formatting is not consistent or correct.	None or unrelated	___ × **0.5**
Overall Style, Organization, and Writing Quality	Appropriate as a piece of scientific or engineering writing, words were chosen carefully and appropriately. Figures of professional quality. Sentence structure was clear and as concise. Includes no excess or unnecessary text. Evidence the report was edited by the author to improve clarity and readability. From a technical standpoint, the paper is free of spelling, punctuation, and grammatical errors	Minimal awkward phrasing or word choices. Report is easy to read and constructed properly. Some excess unneeded text. Evidence of editing. Acceptable quality figures. Few grammar and spelling errors.	Many passages are phrased poorly, contained awkward word choices, or many long sentences. Narrative is disorganized in many places. Substantial excess text that should be eliminated by editing .Tense not appropriate or not in agreement in several places. Multiple grammatical and/or spelling errors.	Poorly organized narrative with frequent awkward phrases and poor word choices. Setences are too long or short. Lacks cohesion, style and fluidity. Many instances of verb tenses not agreeing. No evidence of editing. Frequent spelling and grammar errors. Focus on technical writing improvement critical	Poor style, format, and organization. Extreme writing errors. Report very difficult to read. Focus on technical writing improvement critical	___
					SUM	**/30**

Grading: A-range = 26 to 30 D-range = 13 to 17
B-range = 21 to 26 below 17 = F
C-range = 17 to 21